

MONTAUBAN MORIN-VERDRINES AP, France

WMO: 075400

Lat: 44.0281N

Lon: 1.3767E

Elev: 351

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.8	26.8	17.3	13.3	30.7	21.2	16.1	31.2	23.1	51.1	18.7	49.3	2.7	280	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	93.9	70.4	90.0	69.5	86.8	68.1	72.8	87.5	71.3	85.6	69.9	83.3	6.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	104.5	79.3	66.6	99.4	77.2	65.3	94.8	75.6	36.6	87.4	35.3	86.0	34.0	83.6	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	15.3	12.8	20.1	100.2	4.0	3.8	17.3	102.9	14.9	105.2	12.7	107.3	9.8	110.1
			19.3	75.5	4.0	1.6	16.5	76.7	14.2	77.7	11.9	78.6	9.1	79.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	41.9	43.3	49.8	55.3	61.1	69.3	72.7	71.9	66.1	59.1	48.9	42.8	(6)
	DBStd	12.59	6.47	6.67	5.82	5.30	5.50	6.10	5.26	5.53	5.57	6.72	6.65	6.89	(7)
	HDD50	889	258	198	75	12	1	0	0	0	0	12	97	236	(8)
	HDD65	3690	715	608	472	291	145	26	4	5	52	201	482	689	(9)
	CDD50	3417	8	10	68	172	347	578	702	679	483	292	66	12	(10)
	CDD65	743	0	0	0	2	25	154	241	219	85	17	0	0	(11)
	CDH74	7548	0	0	4	78	322	1547	2415	2181	869	132	0	0	(12)
	CDH80	2959	0	0	0	10	68	598	1036	960	271	16	0	0	(13)
(14) Wind	WSAvg	5.2	4.9	5.7	6.2	6.1	6.0	5.4	5.5	4.9	4.3	4.5	4.7	4.3	(14)
(15) Precipitation	PrecAvg	29.3	2.8	1.9	2.2	3.1	3.1	2.3	2.0	2.3	2.1	2.2	2.9	2.4	(15)
	PrecMax	39.3	5.3	4.2	4.9	6.3	6.1	5.3	5.2	5.7	5.4	3.9	5.9	4.6	(16)
	PrecMin	21.9	0.3	0.3	0.8	0.5	0.8	0.7	0.4	0.5	0.7	0.6	0.6	0.4	(17)
	PrecStd	4.3	1.5	1.1	1.1	1.7	1.4	1.2	1.1	1.2	1.1	0.9	1.3	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.8	65.5	74.3	81.5	85.7	96.8	97.5	100.5	90.7	82.3	70.1	62.0	(19)
		MCWB	54.6	53.2	55.8	61.8	65.5	71.0	70.9	70.8	67.9	65.2	59.8	54.7	(20)
	2%	DB	56.8	61.0	68.7	75.8	81.6	90.3	93.2	94.3	86.7	77.6	66.2	58.3	(21)
		MCWB	52.6	51.5	54.7	59.1	64.5	70.2	70.5	70.1	66.3	64.2	58.1	52.9	(22)
	5%	DB	54.2	57.6	64.7	71.7	77.4	86.4	89.5	89.8	83.0	74.1	62.7	55.5	(23)
		MCWB	51.1	50.0	52.8	57.8	63.2	68.4	69.5	69.5	64.8	62.7	56.3	51.8	(24)
	10%	DB	51.8	54.3	61.2	67.6	73.5	83.0	86.1	85.5	79.2	71.0	59.4	53.1	(25)
		MCWB	49.1	48.7	51.4	55.6	60.8	66.8	68.4	68.2	64.1	61.8	54.8	50.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.0	54.7	58.3	63.3	69.1	74.5	74.4	74.5	71.2	67.9	61.1	56.0	(27)
		MCDB	58.0	60.9	68.0	77.1	80.5	89.3	91.1	90.6	85.8	78.0	67.9	59.6	(28)
	2%	WB	53.8	52.7	56.1	60.8	66.3	72.2	72.7	72.2	69.0	65.9	59.0	54.0	(29)
		MCDB	56.0	58.8	65.4	72.9	78.2	86.5	87.4	87.2	81.7	73.6	64.3	57.1	(30)
	5%	WB	51.6	51.0	54.5	58.6	64.2	70.3	71.2	70.8	67.1	64.3	57.0	52.2	(31)
		MCDB	53.8	56.2	62.6	68.8	74.6	83.5	85.0	85.4	78.6	71.1	61.3	55.0	(32)
	10%	WB	49.5	49.3	52.7	56.6	62.1	68.4	69.6	69.2	65.5	62.7	55.3	50.4	(33)
		MCDB	51.5	53.8	59.5	65.5	71.1	79.9	82.3	82.9	75.8	69.9	59.3	52.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	16.8	18.9	19.3	19.2	20.3	21.0	21.6	22.0	19.6	14.3	13.5	(35)
		MCDBR	16.2	23.5	27.7	28.3	27.3	27.3	28.7	30.7	28.9	24.5	20.0	17.4	(36)
	5% WB	MCWBR	12.0	14.8	15.4	14.1	12.9	10.9	10.2	10.6	11.4	12.7	12.9	12.7	(37)
		MCWBR	13.1	20.0	22.8	24.3	23.1	24.6	24.4	24.9	23.4	19.9	17.1	15.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.320	0.334	0.371	0.400	0.406	0.418	0.410	0.409	0.378	0.365	0.338	0.318	(40)
		taud	2.478	2.419	2.325	2.245	2.282	2.286	2.308	2.321	2.389	2.435	2.462	2.482	(41)
	Ebn at Noon	Ebn at Noon	256	271	274	274	275	271	272	268	268	256	245	244	(42)
		Edh at Noon	23	28	35	42	41	41	40	38	33	28	23	21	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	445	744	1125	1470	1699	1961	1941	1722	1376	904	522	412	(44)
	RadStd	RadStd	47	85	114	168	128	151	141	107	79	73	59	57	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page